

What is Qi?

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Technically, “\$Qi” is a cryptographic token defined by Quai Network, the layer-1 blockchain with several intended purposes which will be discussed in depth in this essay. Philosophically, “Qi” is an older, wider, more ancient concept defined by cultural, historical, and spiritual traditions of human beings observing nature and its natural processes. In this sense, “Qi” represents the “life energy” or “vital flow” observed in nature—that which is deeply embedded in physical and spiritual systems alike. A distinction will be made throughout this essay between \$Qi defined by a blockchain, and Qi defined by nature. This distinction is necessary in order to enable the reader to appropriately switch between ontological and epistemological contexts: one defined entirely by a blockchain with formal, mathematical, deterministic rules, and the other defined entirely by subjective interpretations of world history and subjective experience. Then, when appropriate, blend their mutual reality together to find higher-order relationship. Inevitably, the author believes the token, \$Qi, inherits ideology from human tradition and nature, with intention of representing those same ideas in the realm of the human economy, and it is an appropriate assumption that the engineers of the Quai Network blockchain had the same intention.

In 2026, nearly two hundred and fifty years of America’s semiquincentennial year, the nature of the economy has evolved and has an awareness potential that is capable now of embedding the idea of life energy’s vital flow into our organizing systems. Money, wrought with complexity and land-mines, is a representative system for tracking, facilitating, and organizing vital flows on the scale of our biosphere. To lack a scientific, ontological, and epistemological investigation into our inherited organizing methods would be grossly irresponsible. The question, *what is \$Qi?*, is the pursuit to connect nature and natural processes to economy and economic processes in order to align natural incentives with natural constraints. To say what is occurring in the economy is not natural would be problematic to defend. Instead, I’d like to claim that the economy may have capability to incorporate more awareness about itself—to give it the ability to know what is known and what can’t be known about itself.

Epistemology

Beginning with epistemology, the study of awareness. Typically known as a theory of knowledge, epistemology is reduced to what is and is not known. To know a coinflip is heads is

to know it is not tails. To know it is not tails implies it is heads. Heads = Not Tails, Not Tails = Heads, Not Heads = Tails, Tails = Not Heads. Neither Heads nor Tails = Undetermined, or Vertical Landing.

The structure of that argument—coin lands, outcome is one of two faces, knowledge follows by exclusion—is a particular mode of reasoning, and it is worth naming alongside the others. Epistemology distinguishes three. **Deductive reasoning** moves from a general rule to a specific conclusion. If the coin has two faces and is not heads, it is tails. Given the rule, the conclusion is certain. **Inductive reasoning** moves the other direction, from specific observations to a general pattern. Flip the coin a thousand times, observe roughly even heads and tails, and theorize a fair coin. The conclusion is not certain; it is probabilistic, and another thousand flips could revise it. **Abductive reasoning** begins from an incomplete observation and reaches for the best available explanation. The coin is standing on its edge. We did not predict this; we have no rule that resolves it; we infer—a draft, a breath, a flaw in the surface—and we proceed on the best guess we can make. The conclusion is also probabilistic, but unlike induction it does not rest on a body of prior observation. It rests on what the situation seems to require.

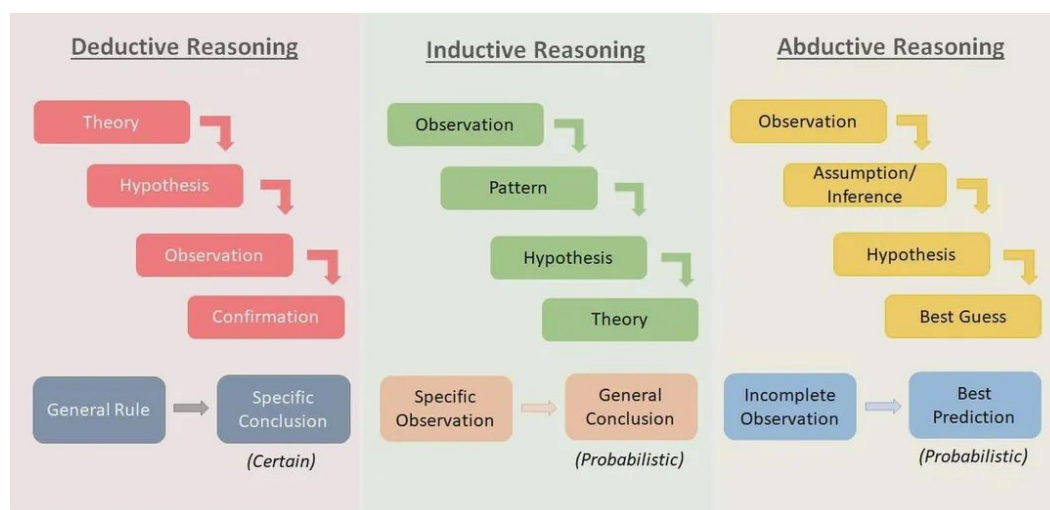


Figure 1. Deductive, inductive, and abductive reasoning. Deduction is certain; induction and abduction are probabilistic.

These three modes are not in competition; they cover different territory. Deduction is the engine of formal systems—mathematics, law, and the deterministic rules of a blockchain. Induction is the engine of empirical science. Abduction is the engine of diagnosis, investigation, design, and most of the decisions a person, a firm, or an economy actually has to make in real time. The Vertical Landing is an abductive event. It is the case deduction cannot resolve and induction has not yet seen enough of to characterize. It demands the third mode.

The Vertical Landing is the interesting case. It is the residue of the binary, the outcome the binary cannot represent, and the case abduction was named to address. A coin that lands on its edge has not failed to produce information; it has produced information the system was not designed to receive. Most working definitions of epistemology presume the system is designed to receive—that knowledge is the set of propositions which resolve cleanly to true or false. But the natural world routinely produces edges. A river is neither the bank nor the sea. A photon is neither here nor there until it is measured. A market is neither in equilibrium nor in collapse; it is in motion. If epistemology is to be useful to a theory of economy, and the economy is to be modeled as a natural process, then the study of awareness has to account for the edges as much as for the faces. It has to hold a third position: what is known, what is not known, and what cannot yet be resolved. This third position is where \$Q_i\$, and the economy that would price it, has to live.

The Wolfram Physics Project

In April 2020, in the suspension of the early pandemic, Stephen Wolfram launched the Wolfram Physics Project. This was a revelation: at a moment with the world at a standstill, one of the world's pioneering computer scientists was launching a fundamental theory of physics and I was his audience. Each day, every day, I watched physicists use the mathematical tools, programs, and software engineering they had built to ask questions about the universe, and I had all the time in the world to learn it.

Within a year, I was hired at Wolfram Research. This was my pitch to Stephen Wolfram for what I wanted to work on:

Computational Irreducibility is the Source of Robust Value in Economics

On a recent interview, Stephen Wolfram said the following: “Computational irreducibility is the source of robust value in economics... in a sense, proof-of-work is a crazy, inner bottle version of that process.” I’d like to explore this idea as deeply as possible, because I believe there is a major connection between computation, energy, and value—which is fundamental to the field of economics and has yet to be fully explored. To make this idea more tangible, this project’s goal would be to map the computational irreducibility of Bitcoin’s proof-of-work mechanism with the total amount of electricity used to secure it. Building this foundational link in Wolfram Language will be the starting point of understanding economic value, by creating a universal metric based on computation and electricity, and from that starting point we can materialize additional layers of economic activity on top of this core model.

Exploration of the Bitcoin blockchain would occur via Wolfram Language functionality. Importing data into WL and visualizing the hash rate of computation, electricity used to perform computation, and economic value currently priced in Bitcoin would be the major sources of data needed. Between current WL functionality and retrieving public blockchain information, I'm sure it's feasible to get the data needed for this project. Then it's just about producing specific metrics, visualizations, and synthesis to offer a theory of economics based on computational irreducibility.

This project is innovative because, as a trained economist, I've always believed there was something missing from my field of study and this is the project I would tackle to complete it. This idea seems possible now for two main reasons: 1) I've only recently become aware of the tools within Wolfram Language and the immense power it offers, and 2) Bitcoin's proof-of-work seems to be a major link between computation, energy, and value and is only 12 years old. Ultimately, the field of economics could use more scientific rigor and I believe this is the direction it should take.

Shortly thereafter I joined Wolfram on the blockchain team as an intern. I read everything Stephen Wolfram and the company wrote or produced, and I began again to earnestly learn computers, programming, and what computation actually is.

For those who don't know, Stephen Wolfram wrote a book in 2001 called *A New Kind of Science* and basically no one has read it from cover to cover. I don't consider that as much an impressive feat as catching up on all of his other content. This is a scientist doing science in public as you would hope scientists would. He built the tools he needed, he uses the tools he built, and he publishes research using those tools. These tools are used by math students around the world, by researchers and scientists around the world, and he eats his own dog food, to say the least.

If one was going to adopt a new framework of physics, I had little to lose by adopting his. So I did.

I'm now five years at Wolfram, the Head of Blockchain Research & Education, and I have my answer to the question for which I came to Wolfram.

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To Be Continued — March 31, 2026